BEING SEPARATED, QUASI-SEPARATED, QUASI-DM OR DM IS INHERITED BY THE FIBRE PRODUCTS OF TEST SPACES AND BY THE ISOM SHEAVES

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Allowed standard prerequisites (editorial).

- The characterisation of separated, quasi-separated, quasi-DM and DM morphisms of algebraic stacks by the corresponding property of the diagonal. Invoked inside the counted proof (source lines 1885-1889). Converts the property of the morphism into a property of the diagonal, which is what the displayed cartesian square tests.
- The description of the Isom sheaf of two objects over a common test space as a fibre product of the double fibre product, and stability of the properties under base change. Used inside the counted proof (source lines 1892-1897). Exhibits the morphism of the second assertion as a base change of the morphism of the first.

Lemma 0.1. *Let $f : \mathcal{X} \to \mathcal{Y}$ be a separated (resp. quasi-separated, resp. quasi-DM, resp. DM) morphism of algebraic stacks. Then*

(1) *given algebraic spaces T_i, $i = 1, 2$ and morphisms $x_i : T_i \to \mathcal{X}$, with $y_i = f \circ x_i$ the morphism*

$$T_1 \times_{x_1, \mathcal{X}, x_2} T_2 \longrightarrow T_1 \times_{y_1, \mathcal{Y}, y_2} T_2$$

is proper (resp. quasi-compact and quasi-separated, resp. locally quasi-finite, resp. unramified),

(2) *given an algebraic space T and morphisms $x_i : T \to \mathcal{X}$, $i = 1, 2$, with $y_i = f \circ x_i$ the morphism*

$$Isom_{\mathcal{X}}(x_1, x_2) \longrightarrow Isom_{\mathcal{Y}}(y_1, y_2)$$

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is proper (resp. quasi-compact and quasi-separated, resp. locally quasi-finite, resp. unramified).

Proof. Proof of (1). Observe that the diagram

$$\begin{array}{ccc} T_1 \times_{x_1,\mathcal{X},x_2} T_2 & \longrightarrow & T_1 \times_{y_1,\mathcal{Y},y_2} T_2 \\ \downarrow & & \downarrow \\ \mathcal{X} & \longrightarrow & \mathcal{X} \times_{\mathcal{Y}} \mathcal{X} \end{array}$$

is cartesian. Hence this follows from the fact that f is separated (resp. quasi-separated, resp. quasi-DM, resp. DM) if and only if the diagonal is proper (resp. quasi-compact and quasi-separated, resp. locally quasi-finite, resp. unramified).

Proof of (2). This is true because

$$Isom_{\mathcal{X}}(x_1, x_2) = (T \times_{x_1,\mathcal{X},x_2} T) \times_{T\times T,\Delta_T} T$$

hence the morphism in (2) is a base change of the morphism in (1). □